

IT Helpdesk Ticket Management

System: Database Design Proposal

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Course: CPSC 332: File Structures and Database Systems

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System Description:

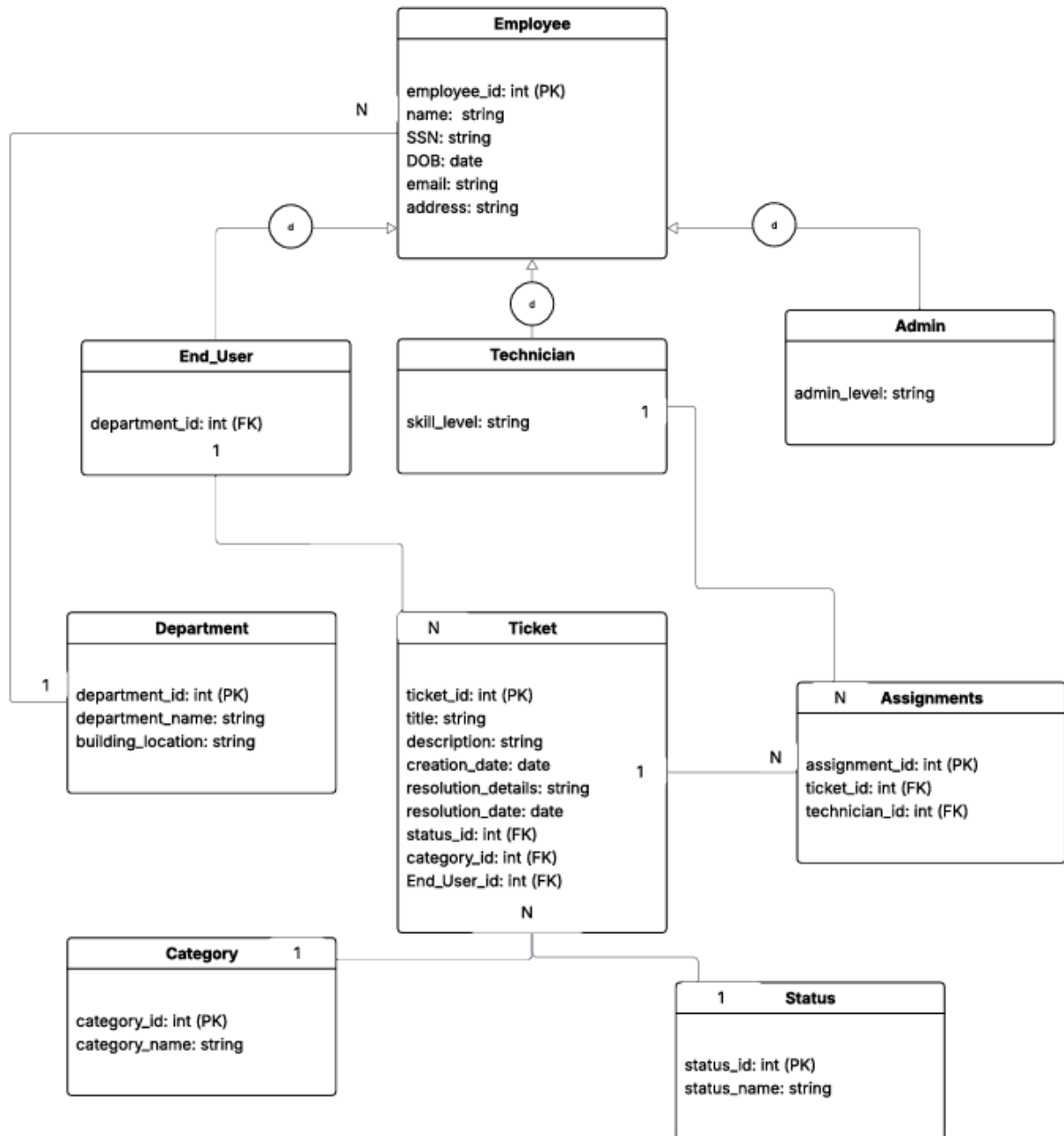
The IT Helpdesk Ticket Management System is a relational database designed to support tracking, management, and resolution of IT support requests within an organization. This system centralizes all help desk operations by allowing users to submit tickets, IT staff to manage and resolve issues, and administrators to monitor performance and service efficiency.

The system is designed to be used by employees (end users), IT support technicians, and system administrators. It will track data such as user information, submitted tickets, assigned technicians, ticket status, and resolution history. The workflow will include ticket submissions, assignments, status updates, and resolution logging. This system ensures accountability, efficient response times, and enables data-driven decision making for IT support services.

Scope:

- The system will allow employees to submit IT support tickets.
- The system will store employee account information, roles, and department.
- The system will assign tickets to IT technicians.
- The system will track ticket status throughout its lifecycle.
- The system will record ticket resolution details and timestamps.
- The system will categorize tickets based on issue type.
- The system will maintain a history of ticket updates.
- The system will prevent duplicate or invalid ticket entries.

UML Overview (High-level)



Business Rules / Constraints

1. Each employee must have a unique employee_id, email, and SSN.
2. Each employee must belong to exactly one subclass: End_User, Technician, or Admin.
3. Each employee must belong to exactly one department.
4. Only End_Users are allowed to submit tickets.
5. A ticket must be associated with exactly one End_User.
6. A ticket can be assigned to multiple Technicians, but assignments must exist in the Assignments table.
7. A Technician must exist before being assigned to a ticket.
8. Ticket status must be one of {Open, In Progress, Closed}.
9. A ticket resolution can only exist if the ticket is marked as “Closed.”
10. A ticket's resolution date must be later than its creation date.
11. Employee DOB must be a valid past date.
12. Duplicate tickets (same user, same title, same timestamp) are not allowed.